

Prescribed burn severity has minimal effect on common bird species in a fire-prone forest ecosystem

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ABSTRACT

Prescribed burning is widely used to mitigate the effects of severe fires across the landscape and to maintain biodiversity. Just like wildfires, the severity of prescribed burns can vary; this study was an opportunistic investigation. In one fortnight during autumn months of 2012, several prescribed burns were carried out in heathy-dry forests of central Victoria, Australia. We used measurements of canopy scorch, bark burn and ground cover burn to calculate a severity score for each site. The severity scores across sites ranged from low (2.5) to high (10). A before-after control-impact (BACI) design was utilised to model the potential impacts of fire and fire severity on birds. We used generalised linear mixed models (GLMM's), and incorporated first- and second-year post-fire spring/summer observations from 2012 to 2014, against bird data from observations carried out in 2010. The total combined abundances of individual species showed that broadly, bird abundance rebounded to pre-burn levels by the second spring post-fire. There was little response detected in either species richness or turnover. The muted turnover result aligns to other studies that indicate a scarcity of early-successional-stage species in eucalyptus forests and woodlands that rapidly regenerate post-fire. Ten individual species were also examined, and only one species, the White-throated Treecreeper (*Cormobates leucophaea*), responded to both fire and its severity. The BACI design was informative in illustrating that while the forest birds were resilient to small-scale prescribed burns of any severity, abundances in general may have been in decline, a result aligning with the years of reduced rainfall in the region.

1. Introduction

Fire is a global driver of change to ecosystem states and functions, shaping the distribution and abundance of plant and animal species (Bond and Keeley, 2005; Pausas and Keeley, 2009; Bowman et al., 2016; He and Lamont, 2017; He et al., 2019). As global climate systems increase in variability, fire regimes change, resulting in complex changes to vegetation around the globe (Pausas and Vallejo, 1999), with consequences for forest biodiversity (Gil-tena et al., 2009). Research analysing the impact of climate change on fire predicts an increase in forest fire frequency (Cary, 2002; Pitman et al., 2007). For example, in Australia, this increase has been realized within the last decade with mega-fires increasing in frequency (Fairman et al., 2016). This trend is predicted to continue (Pitman et al., 2007; Steffen et al., 2017), and has resulted in calls to increase prescribed burning in efforts to mitigate wildfire impacts on both human life and assets (Teague,

2010). Prescribed burning is the controlled use of fire to achieve management goals and is used globally as a means of managing fire fuel loads: across southern Europe (Fernandes et al., 2013); in South African savanna ecosystems (Brockett et al., 2001); in North America (Artman et al., 2005; Brudvig et al., 2007) and in Australia (Teague, 2010). Hence, it is important to understand the responses of biodiversity to prescribed fire regimes.

Faunal responses to fire are often interpreted using principles of secondary succession (Noble and Slatyer, 1980), as there is a link between the temporal changes to vegetation structure and the abundance of resources for fauna (Friend, 1993; Haslem et al., 2011; Swan et al., 2015; Kelly et al., 2017). Although the term succession has been criticised for its ambiguity (Binkley et al., 2015), it provides a useful framework for evaluating how the occurrence or abundance of fauna varies as vegetation regenerates after fire (Di Stefano et al., 2013). Moreover, successional stages often form the basis of ecological fire

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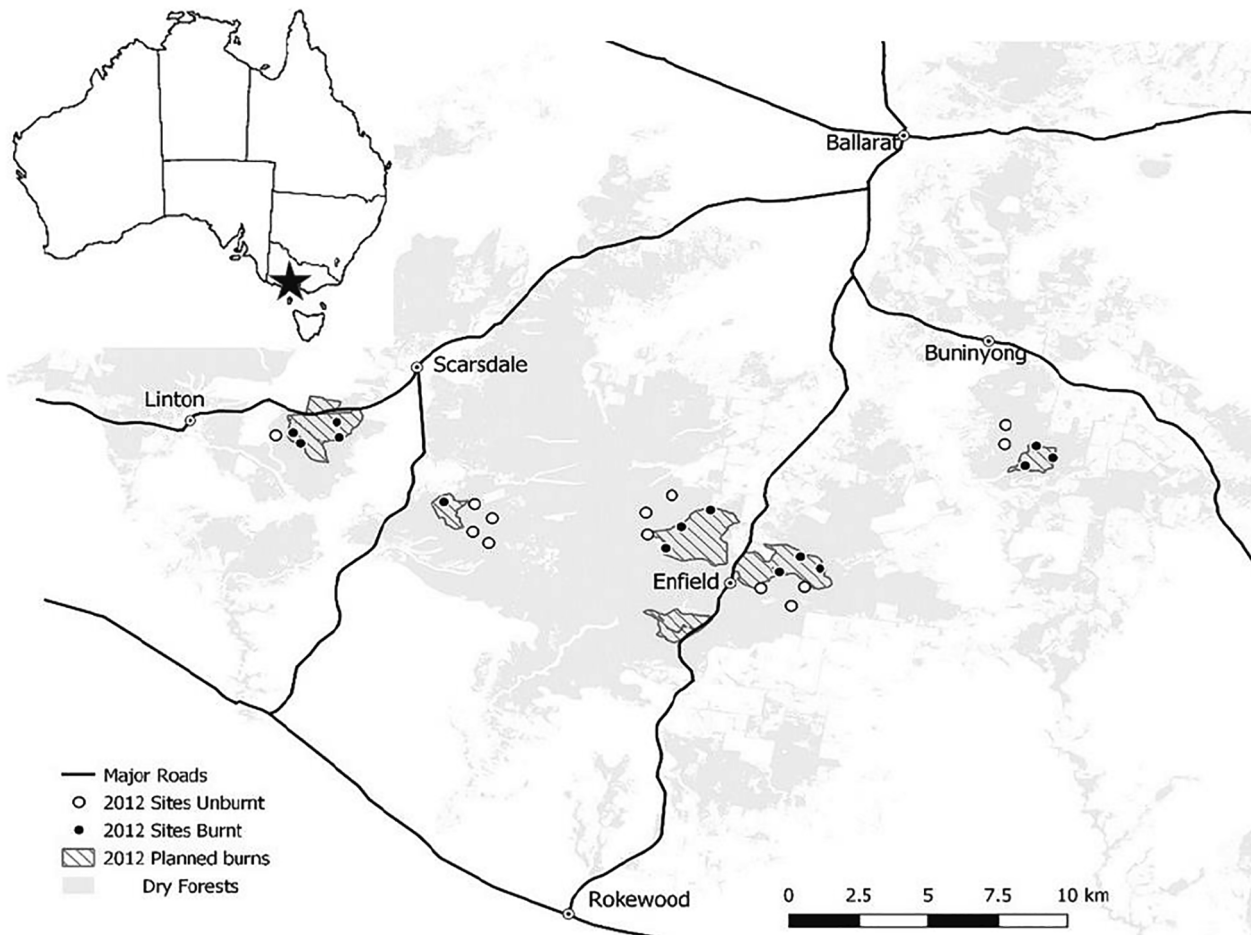


Fig. 1. Site locations in Victoria, Australia relative to prescribed burns of 2012.

management because they are easier to map and measure than other fire-regime variables, and different species resource requirements may be met by different successional stages (Sitters et al., 2018, Swan et al., 2018). For example, studies on birds from around the globe show that changes in fire-mediated vegetation structure have the capacity to alter the composition of avian communities (Fontaine et al., 2009, Albanesi et al., 2012, Sitters et al., 2015). In some cases however, fire may have little effect, presumably because important resources are still present in the post-fire environment (Loyn and McNabb, 2015) and birds may be influenced by factors other than fire (Woinarski and Recher, 1997). However, birds may be highly responsive to spatiotemporal environmental change caused by fire and other forms of disturbance (Loyn and McNabb, 2015, Leonard et al., 2016, Kelly et al., 2017).

While literature on the avifaunal responses to fire is increasing, some argue that there is a paucity of information on how animals respond to fire severity (Kotliar et al., 2007, Lindenmayer et al., 2013). The term 'fire severity' was borne from the need to provide a description on how a fire has affected ecosystems, with a common above-ground metric being measurements of the fire-induced changes to vegetation; loss of organic matter, such as canopy and trunk-bark scorch and surface ground vegetation or litter (Monamy and Fox, 2000, Kotliar et al., 2007, Keeley, 2009, Lindenmayer et al., 2014, Espinosa et al., 2019). Fire severity is an aspect of the fire regime that has a major influence on post-fire vegetation dynamics, and hence species responses (Smucker et al., 2005, Kotliar et al., 2007, Fontaine and Kennedy, 2012, Lindenmayer et al., 2014, Loyn and McNabb, 2015).

In Australia, while landscape-level high-severity fire may have a strong effect on bird species richness (Loyn, 1997, Lindenmayer et al., 2014), various studies have shown that there are few early-successional

species adapted to recently-burnt environments because eucalypt forests regenerate rapidly after fire (Loyn, 1980, Lindenmayer et al., 2008, Loyn and McNabb, 2015, Sitters et al., 2014a, Hutto et al., 2015, Sitters et al., 2015). The bulk of biota are found to be resilient to fire due to *in situ* landscape factors (Bradstock, 2008), especially when areas of unburnt vegetation remain within the burnt area (Robinson et al., 2014). Where vegetation changes dramatically post-fire, so too may the bird community change. In a mulga woodland of the central Australian arid zone, an early-successional stage was one of grasslands, with a bird community largely comprising granivores, distinctly different from the older, long-unburnt vegetation nearby largely comprising foliar insectivores (Leavesley et al., 2010). However, changes may not be consistent across a landscape. In the north-eastern section of the Iberian Peninsula, bird assemblages on sites immediately post-fire varied across the region (Brotons et al., 2005), reflecting variation in the ability of species to disperse and colonise habitats post-wildfire. Temperate eucalypt forests are listed as one of the ten Australian ecosystems most vulnerable to tipping points, where environmental changes (e.g. changing fire regimes, increased temperatures, habitat fragmentation) may result in an abrupt shift to a contrasting ecosystem state (Laurance et al., 2011). As a consequence, animals living in temperate forest systems are at risk (Department of Environment Land Water and Planning, 2017).

While the response of birds to fire has been studied across many Australian regions, studies in south-east Australia have, to date, been focused on foothill forests that include wet gullies (Loyn, 1997, Loyn et al., 2003, Robinson et al., 2014, Sitters et al., 2014a, Sitters et al., 2014b, Sitters et al., 2015, Loyn and McNabb, 2015, Haslem et al., 2016, Leonard et al., 2016, Kelly et al., 2017); the taller and wetter

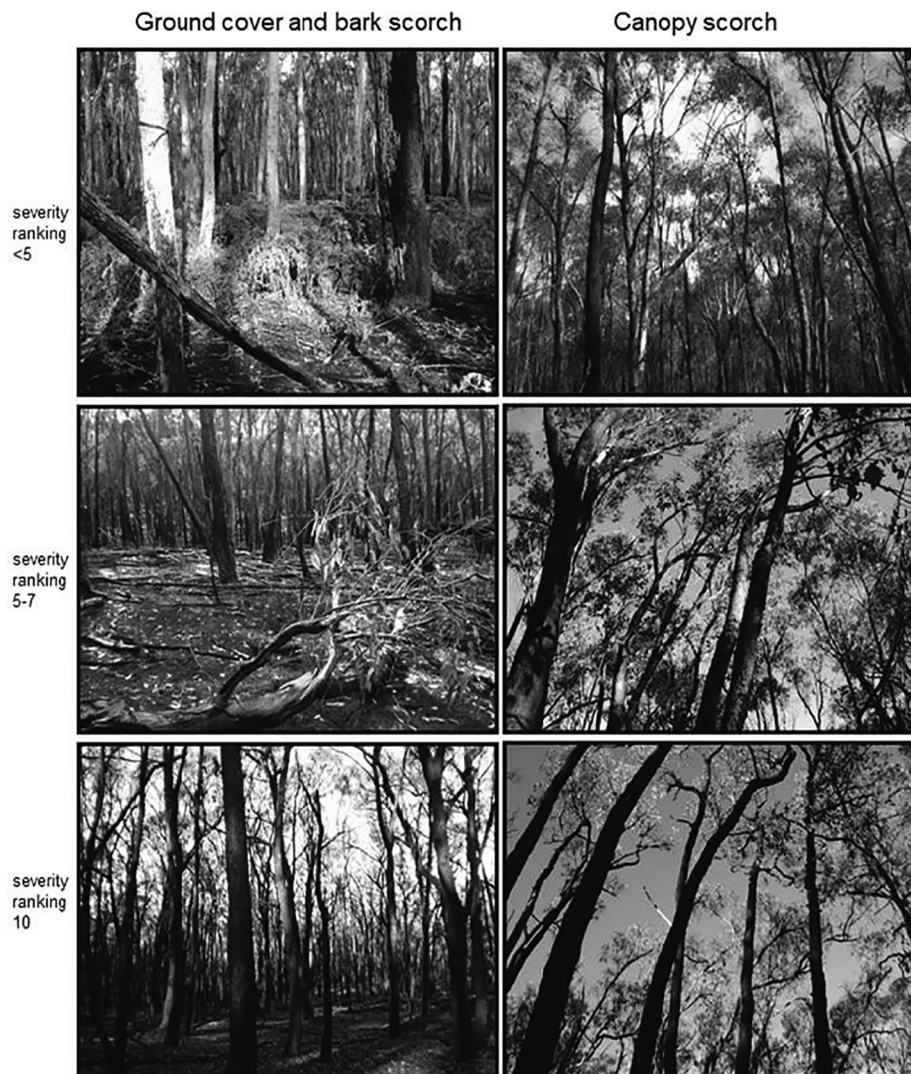


Fig. 2. The severity of prescribed burns, illustrated with examples of ground cover, bark and canopy scorch on burn sites. The tree trunks in the images with severity ranking of 10 were entirely blackened and canopy leaves all subsequently fell to the ground. However, most of the trees survived and regenerated from epicormic shoots.

mountain ash (*Eucalyptus regnans*) forests in central Victoria (Lindenmayer et al., 2014); and the semi-arid regions including mallee (Brown et al., 2009, Taylor et al., 2012, Taylor et al., 2013, Watson et al., 2012).

Our study focuses on heathy-dry forests, which are widespread in the lowlands of central and western Victoria, where fire management is often complicated by interspersed people and native vegetation.

We identified an opportunity to study effects of prescribed fire on birds in heathy-dry forests, using a before-after control-impact (BACI) design (Stewart-Oaten et al., 1986). Bird surveys had already been conducted in 2010 before a program of prescribed burns by the Department of Environment, Land, Water and Planning (DELWP) was carried out in 2012. Alpha and beta diversity, along with individual species' responses to prescribed burn severity, were measured. We anticipated that after severe burns, bird species richness would initially decrease, followed by a rapid recovery. Rapid vegetation regrowth post-fire may influence bird assemblages, so we also predicted an increase in species turnover in the first year post-fire, and low turnover relative to the pre-fire level, in the second year post-fire. Lastly, we anticipated that individual species' responses to fire severity would reflect the degree to which essential resources were removed, for those species that display nest or site fidelity.

2. Materials and methods

2.1. Study area

The study was conducted in the Central Victorian Uplands bioregion in Victoria, Australia (Fig. 1). The region has a temperate climate with a winter mean monthly temperature range of 3.2 °C (daily minimum) to 13.8 °C (daily maximum). The equivalent mean monthly summer temperature range is 8.8–27.8 °C (Bureau of Meteorology, 2017). Rainfall recorded at Durham Lead (the centre of the study area) is representative of the region with an annual mean of 830.5 mm from 2009 to 2012 (Bureau of Meteorology, 2017).

Dry, temperate eucalypt forests comprise the major natural vegetation that remains in the study area. All sites selected for this study were within vegetation classified as Grassy/Heathy Dry Forests (Cheal, 2010). This forest type supports a low, relatively open understorey dominated by tussock grasses (Poaceae), with low- to medium-height sclerophyllous shrubs (e.g. Epacridaceae, Fabaceae). The eucalyptus trees, with maximum 30% percent canopy cover and a characteristic height of 25 m, include stringybark (e.g. Messmate *Eucalyptus obliqua*, Brown Stringybark *E. baxteri*) and peppermint species (Narrow-leaved Peppermint *E. radiata* and Broad-leaved Peppermint *E. dives*), as well as

scattered gum-barked species such as Manna Gum (*E. viminalis*).

Wildfires in this region occurred around Linton in 1939, Durham Lead in 1972 and Enfield in 1995, while prescribed burns have been applied broadly across the landscape to achieve fuel-reduction goals (Department of Sustainability and Environment, 2004), (Fig. 1), (Supplementary S1).

2.2. Site selection and fire treatment

Twenty-seven sites were selected in 2010 within large, intact areas of native vegetation (Fig. 1). Each site was two hectares in size, at least 600 m from neighbouring sites, a minimum of 50 m from roads and burn edges, and at least 100 m from drainage lines and major clearings.

Prescribed burns occurred during a two-week period in March and April of 2012, affecting 14 sites (Fig. 1). The burns were conducted by the local land management agency, DELWP and were part of a fire program designed to reduce fuel loads. All control sites had not burned within the eight years prior to conducting bird observations (Supplementary S1).

2.3. Burn severity

Canopy scorch, bark burn and ground cover were visually assessed within three months of fire, at all burnt sites. The extent to which these three parts of the vegetation burnt, resulted in a percentage-based impact score for each burn site.

Each site was a 100 m by 200 m rectangle, with steel pegs at each corner and at the middle of each side. A ten-metre-diameter circle was measured around each of eight pegs. Within each circle, the following parameters were visually assessed by one observer and given a score out of 100 (with 100 being completely burnt or completely bare and zero being unburnt): canopy cover, tree charring, vegetation cover > 1 m tall, vegetation cover < 1 m tall, fine litter (decomposing), leaf cover (ground), bare ground, grasses up to 30 cm, tussock and herb presence. Scores for each site were pooled, equally weighted and the final total then transformed to a score out of 10. Severity scores ranged from 2.5 to 10, in terms of representing loss of green vegetation and fine fuel (Fig. 2) (Supplementary S2). The score of 10 implies that the entire site had experienced fire severe enough to remove green vegetation and fine fuel, though generally most trees had survived.

2.4. Bird data

The 20-minute search method (Loyn, 1986) was used to survey birds. Records were made of birds seen and heard while walking across the site over a 20-minute period. Birds flying overhead, or heard and considered to be outside the plot, were not recorded. All surveys were undertaken in mild weather with no monitoring undertaken on days of extreme heat (> 30 °C) or when wind was greater than 20 km/h.

Birds were surveyed on all sites twice before the fires, between September 2010 and February 2011 (spring/summer), twice six months after the fire between September 2012 and February 2013, and twice 18 months after the fire between September 2013 and February 2014. The pairs of surveys conducted at the three time periods were combined, resulting in a single sample representing the bird community before the fire and at six and 18 months (the first and second breeding seasons) after the fire. Equal numbers of morning and afternoon monitoring sessions were carried out for each site.

The pre-burn monitoring that occurred in the spring/summer of 2010 formed part of an independent project. There were two different observers for this project; one collected pre-burn data, the second collected post-burn data. However, while there may be observer differences, this does not affect inferences about fire effects using a BACI model. The BACI model involves a comparison of the mean before-after difference at control sites and the mean before-after difference at treatment sites, and then calculates the difference between these

changes, to determine the interaction effect. So, any observer effect will be consistent across control and impact sites.

2.5. Statistical methods

Generalised linear mixed models (GLMMs) were used to examine the responses of birds to fire and the severity of prescribed burns. Mixed effects models were necessary to account for a nested data structure due to repeated sampling through time and clustering of sites around the location of prescribed burns (Zuur et al., 2009). For each response variable separately (see below), Akaike's Information Criterion, corrected for small sample size (AICc) (Akaike, 1973), was used to determine the most parsimonious random effect structure following the approach outlined in Zuur et al. (2009). Random-effect structures tested included cluster (a factor identifying groups of sites clustered around each prescribed burn), site and site nested within a cluster. In each case the random effect structure with the lowest AICc was used in the analyses described below.

Three response variables were used to assess the effect of fire (burnt, unburnt) and fire severity on the bird community: total bird abundance, species richness and beta diversity. Beta-diversity estimates were derived from presence-absence data and represented species turnover between the (a) pre-fire and six-months post-fire and (b) pre-fire and 18-month post-fire time periods. Estimates were based on the Sørensen dissimilarity index, which ranges from zero (complete similarity) to one (complete dissimilarity) (Baselga and Orme, 2012). In addition, the response of selected species to fire and fire severity was assessed using presence-absence data. Of the 51 species recorded, 10 were observed frequently enough to model. In preliminary analysis, models were run using an abundance index based on total counts and obtained similar results. Results of the presence-absence models are reported here.

Modelling was conducted in two stages. In stage one the response of birds to fire was modelled using two categorical predictor variables: fire (burnt, unburnt) and time (before fire, first breeding season after fire, second breeding season after fire). For each response variable we used AICc to compare an additive model (fire + time) to a model including the fire-by-time interaction (fire × time). A fire response was inferred if the model including the interaction was supported by the data.

In stage two the response of birds to fire severity was modelled using one continuous predictor variable representing fire severity (range 0–10) and the categorical predictor time, as described above. For each response variable AICc was used to compare three models: fire severity, fire severity + time and fire severity × time. As above, a fire response was inferred if the model including the interaction was supported by the data.

We calculated power to detect an interaction between fire and time (stage one) and severity and time (stage two) using a subset of response variables (Supplementary S3). Our sample sizes (stage one: $n = 154$; stage two: $n = 84$) provided sufficient statistical power to detect medium and large effects (83–99% power), but not small effects (45–57%). (Results and the range of effect sizes we tested are provided in Supplementary S3).

For analyses involving abundance and species richness, GLMMs were applied with a Poisson distribution. If models were overdispersed the inclusion of an individual-level random effect reduced overdispersion, in all cases (Xavier, 2014). A Gaussian distribution was used for beta diversity models and a binomial distribution was used for models based on presence-absence data. In the case of Poisson and Gaussian models, residual plots (a fitted-values plot and a histogram of residuals) indicated that model assumptions had been met (Zuur et al., 2009). Model fit was assessed using marginal R^2 (variance explained by fixed factors) and conditional R^2 (variance explained by the whole model) (Nakagawa and Schielzeth, 2013).

All analyses were conducted in the R statistical environment (R Development Core Team, 2008), using the following packages: betapart (Baselga and Orme, 2012) for calculating beta diversity; lme4 (Bates

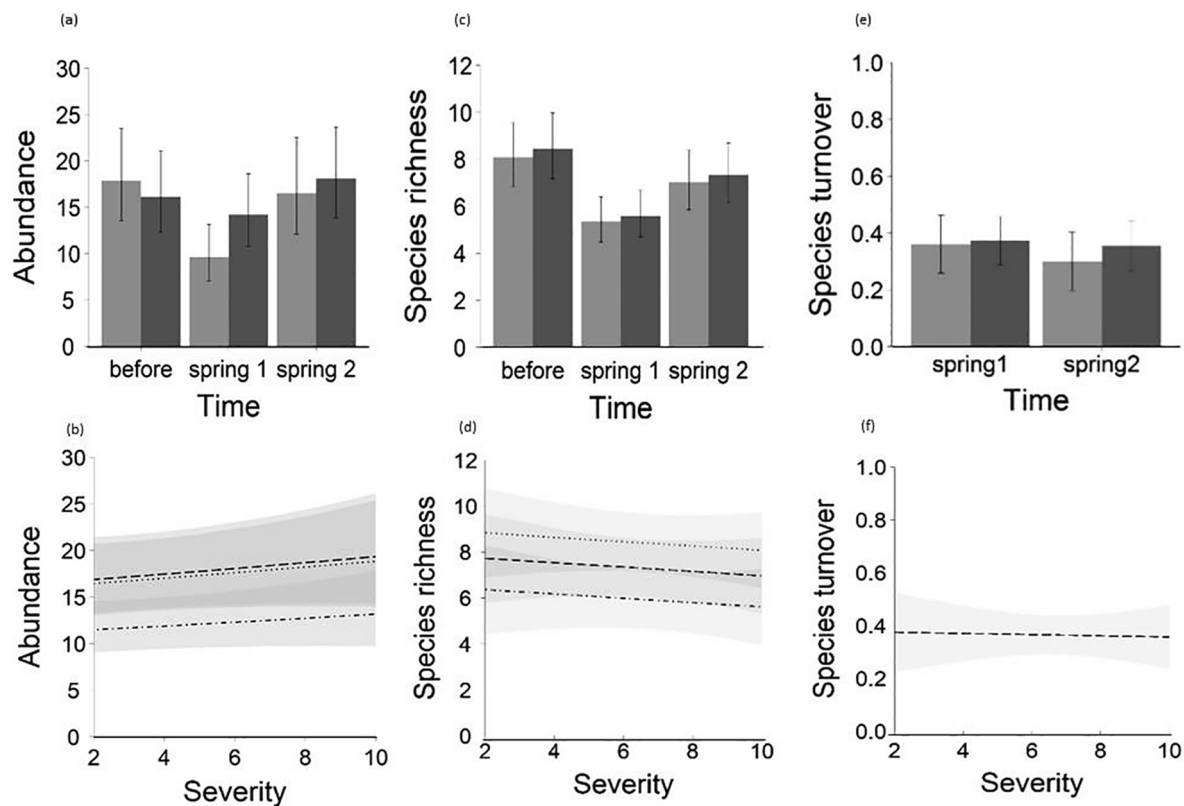


Fig. 3. Avian community responses to fire and the severity of prescribed burns. Community abundance (a, b), species richness (c, d) and species turnover (e, f). Fire graphs have pairs of control sites (light grey) with burn sites (dark grey) for time periods before the autumn prescribed burn (before) followed by the first observation year (spring1) and the second observation year (spring2). Species turnover graphs are a Sørensen dissimilarity measure of species turnover between before and spring1 compared with before and spring2. The fixed effect of time is illustrated with a baseline for before (dotted line) on the burnt sites with comparisons for the two years post fire: spring1 (dot and dash) and spring2 (long dash). Graphs show predicted mean effects from generalized linear mixed models and grey shading/error bars represent 95% confidence limits.

et al., 2016), nlme (Pinheiro et al., 2016) and gamm4 (Wood and Scheipl, 2017) for fitting mixed effects models; MuMIN (Barton, 2016) for model selection and R^2 calculation; and ggplot2 (Wickham and Chang, 2016) for graphing.

3. Results

Fifty-one species were detected and 2825 birds were counted in total (Supplementary S4).

3.1. Alpha- and beta-diversity responses to fire and the severity of prescribed burns

Total bird abundance decreased during the first year post fire across both burn and control sites, and the decrease was greater in the control sites. By the second breeding season, post-fire community abundance levels had rebounded to levels similar to pre-burn abundance, across both burnt and unburnt sites (Fig. 3a, Table 1). We were unable to detect any significant impact of fire severity on bird abundance (Fig. 3b, Table 2).

Bird species richness was not influenced by prescribed fire (Fig. 3c, Table 1). Post-fire changes in species richness were reflected in both burnt and unburnt sites (Fig. 3c) and species richness changed little with an increase in burn severity (Fig. 3d, Table 2).

There was no evidence of a species turnover response to prescribed fire or the severity of the burns (Fig. 3e-f, Tables 1, 2).

3.2. Individual species' responses to prescribed fire and the severity of prescribed burns

Occurrence of the White-throated Treecreeper (*Cormobates leucophaea*) increased in the first spring post-fire, followed by a decrease in the second spring (Fig. 4a). Although the model including the fire interaction was highest-ranked, the weight was only 0.57 and it was not statistically significant (Table 1). However, there was a significant negative effect of severity on treecreeper occurrence in the second spring post-fire (Table 2, Fig. 4b).

Models of occurrence for the remaining nine species showed no detectable fire or severity responses. The occurrence of five species (Buff-rumped Thornbill (*Acanthiza reguloides*), Crimson Rosella (*Platycercus elegans*), Grey Currawong (*Strepera versicolor*), Spotted Pardalote (*Pardalotus punctatus*) and Striated Thornbill (*Acanthiza lineata*)) showed similar decreases in occurrence changes post-fire across both burnt and unburnt sites. Furthermore, there was a limited response to fire severity, with no temporal variations evident (example: Fig. 4c, d). The occurrence of the remaining four species: Grey Fantail (*Rhipidura albiscapa*), Grey Shrike-thrush (*Colluricincla harmonica*), Rufous Whistler (*Pachycephala rufiventris*) and Scarlet Robin (*Petroica boodang*) generally increased over time, but temporal changes were similar at burnt and unburnt sites (example: Fig. 4 e, f).

Individual species' models were also run using abundance data and results were similar to those of presence-absence data. Results for the abundance models for the Crimson Rosella, Grey Currawong, and Rufous Whistler reflected the community fire abundance trends in decreasing after fire and rebounding by the second breeding season (Supplementary S5).

Table 1

Avian responses to fire. Community: abundance, species richness and species turnover; species: probability of occurrence. Predictor variables were fire (burnt, or unburnt), and time (before fire, first breeding season after fire, second breeding season after fire). Models ranked using Akaike's Information Criterion (AICc). Top-ranked models are presented; model sets comprised fire and time in additive and interactive combination. Fixed effect of time reported as two measures: first year post fire (spring1) and second year post fire (spring2) with parameter estimates, standard errors (SE) and statistical significance (P). The intercept is the reference level in the model, with positive estimates indicating an increase above the intercept mean. Marginal R² (R²(m); fixed effects) and conditional R² (R²(c); fixed plus random effects) give the model fit. Predictions are from generalized linear mixed models.

Response variable	Model	Akaike weight		Estimates	SE	P	R ² (m)	R ² (c)
Total bird abundance	fire × time	0.75	intercept	2.86	0.09	< 0.001	0.30	0.44
			fire	0.12	0.12	0.34		
			spring1	−0.02	0.06	0.78		
			spring2	0.16	0.06	0.01		
			fire × spring1	−0.46	0.09	< 0.001		
			fire × spring2	−0.17	0.09	0.08		
species richness	fire + time	0.84	intercept	2.13	0.08	< 0.001	0.17	0.27
			fire	−0.04	0.06	0.48		
			spring1	−0.41	0.07	< 0.001		
			spring2	−0.14	0.07	0.05		
species turnover	fire + time	0.74	intercept	0.35	0.05	0.00	0.02	0.64
			fire	0.03	0.06	0.59		
			time	0.03	0.03	0.21		
Buff-rumped Thornbill <i>Acanthiza reguloides</i>	fire + time	0.89	intercept	0.69	0.38	0.07	0.06	0.09
			fire	−0.30	0.35	0.40		
			spring1	−0.63	0.40	0.12		
			spring2	0.41	0.43	0.34		
Crimson Rosella <i>Platycercus elegans</i>	fire + time	0.87	intercept	−0.09	0.32	0.79	0.04	0.04
			fire	0.45	0.33	0.17		
			spring1	−0.08	0.39	0.85		
			spring2	0.62	0.42	0.14		
Grey Currawong <i>Strepera versicolor</i>	fire + time	0.88	intercept	−0.47	0.40	0.24	0.04	0.10
			fire	−0.15	0.37	0.69		
			spring1	−0.35	0.42	0.40		
			spring2	0.59	0.42	0.16		
Grey Fantail <i>Rhipidura albiscapa</i>	fire + time	0.82	intercept	−1.67	0.49	0.00	0.07	0.24
			fire	0.42	0.50	0.41		
			spring1	1.02	0.46	0.03		
			spring2	0.99	0.48	0.04		
Grey Shrike-thrush <i>Colluricincla harmonica</i>	fire + time	0.88	intercept	−1.11	0.47	0.02	0.06	0.16
			fire	0.04	0.38	0.91		
			spring1	0.19	0.44	0.66		
			spring2	1.10	0.44	0.01		
Rufous Whistler <i>Pachycephala rufiventris</i>	fire + time	0.90	intercept	−0.84	0.46	0.07	0.09	0.19
			fire	0.29	0.40	0.47		
			spring1	−1.07	0.48	0.03		
			spring2	0.28	0.43	0.52		
Scarlet Robin <i>Petroica boodang</i>	fire + time	0.80	intercept	−0.47	0.36	0.19	0.04	0.09
			fire	−0.33	0.37	0.37		
			spring1	0.33	0.41	0.42		
			spring2	0.84	0.42	0.05		
Spotted Pardalote <i>Pardalotus punctatus</i>	fire + time	0.59	intercept	0.24	0.33	0.46	0.08	0.08
			fire	0.57	0.33	0.09		
			spring1	−1.00	0.40	0.01		
			spring2	−0.84	0.41	0.04		
Striated Thornbill <i>Acanthiza lineata</i>	fire + time	0.72	intercept	1.83	0.47	< 0.001	0.09	0.13
			fire	−0.11	0.39	0.78		
			spring1	−1.41	0.48	0.00		
			spring2	−0.85	0.50	0.09		
White-throated Treecreeper <i>Cormobates leucophaea</i>	fire × time	0.57	intercept	−0.15	0.39	0.70	0.18	0.18
			fire	0.15	0.55	0.78		
			spring1	1.15	0.59	0.05		
			spring2	2.35	0.84	0.01		
			fire × spring1	0.64	0.88	0.47		
			fire × spring2	−1.60	1.00	0.11		

4. Discussion

We investigated bird responses to a fire-severity gradient from prescribed burns, and found that total bird abundance was reduced in the first year post-fire, but it rebounded by the second year. There was

no evidence of a community response to the severity of the burns. Of the ten individual species modelled, only one (White-throated Treecreeper) showed a significant burn-severity response, declining in the second year to a greater extent after severe burns than after moderate burns. We discuss the insights arising from this work and their

Table 2

Avian responses to the severity of prescribed burns. Community: abundance, species richness and species turnover; species: probability of occurrence. Predictor variables were fire (burnt, or unburnt), and time (before fire, first breeding season after fire, second breeding season after fire). Models ranked using Akaike's Information Criterion (AICc). Top ranked models reported as Akaike weights from three potential candidates: severity alone and in additive and interactive combination with time. Fixed effect of time reported as two measures relative to the reference level (before fire): first year post fire (spring1) and second year post fire (spring2) with parameter estimates, standard errors (SE), and statistical significance (P). The intercept is the reference level in the model, with positive estimates indicating an increase above the intercept mean. Marginal R^2 ($R^2(m)$; fixed effects) and conditional R^2 ($R^2(c)$; fixed plus random effects) give the model fit. Predictions are from generalized linear mixed models.

Response variable	Model	Akaike weight		Estimates	SE	P	$R^2(m)$	$R^2(c)$
Total bird abundance	severity + time	0.68	intercept	2.41	0.13	< 0.001	0.07	0.88
			severity	0.02	0.02	0.28		
			spring1	0.38	0.13	0.003		
			spring2	0.36	0.12	0.004		
species richness	severity + time	0.75	intercept	2.21	0.16	< 0.001	0.12	0.28
			severity	-0.01	0.02	0.46		
			spring1	-0.35	0.12	0.004		
			spring2	-0.14	0.11	0.22		
species turnover	severity + time	0.69	intercept	0.38	0.10	0.003	0.00	0.79
			severity	0.002	0.01	0.88		
			time	0.02	0.02	0.48		
Buff-rumped Thornbill <i>Acanthiza reguloides</i>	severity	0.69	intercept	0.99	0.63	0.11	0.02	0.02
			severity	-0.11	0.09	0.22		
Crimson Rosella <i>Platycercus elegans</i>	severity	0.63	intercept	0.18	0.76	0.81	0.01	0.06
			severity	0.05	0.11	0.63		
Grey Currawong <i>Strepera versicolor</i>	severity	0.53	intercept	-0.64	1.02	0.54	0.00	0.08
			severity	0.01	0.15	0.92		
Grey Fantail <i>Rhipidura albiscapa</i>	severity + time	0.62	intercept	-1.39	1.01	0.17	0.09	0.25
			severity	-0.01	0.13	0.91		
			spring1	1.49	0.65	0.02		
			spring2	1.15	0.65	0.07		
Grey Shrike-thrush <i>Colluricincla harmonica</i>	severity + time	0.58	intercept	-2.50	1.53	0.10	0.12	0.38
			severity	0.20	0.19	0.29		
			spring1	< 0.001	0.63	1.00		
			spring2	1.25	0.62	0.04		
Rufous Whistler <i>Pachycephala rufiventris</i>	severity + time	0.49	intercept	0.21	0.84	0.81	0.11	0.14
			severity	-0.13	0.11	0.25		
			spring1	-0.98	0.65	0.13		
			spring2	0.32	0.56	0.57		
Scarlet Robin <i>Petroica boodang</i>	severity + time	0.53	intercept	0.38	0.74	0.61	0.17	0.19
			severity	-0.24	0.10	0.02		
			spring1	0.89	0.61	0.15		
			spring2	1.36	0.62	0.03		
Spotted Pardalote <i>Pardalotus punctatus</i>	severity	0.83	intercept	-0.20	0.61	0.74	0.01	0.01
			severity	0.06	0.09	0.49		
Striated Thornbill <i>Acanthiza lineata</i>	severity	0.59	intercept	1.49	0.69	0.04	0.02	0.02
			severity	-0.09	0.09	0.32		
White-throated Treecreeper <i>Cormobates leucophaea</i>	severity × time	0.99	intercept	-1.26	1.16	0.28	0.38	0.41
			severity	0.19	0.16	0.24		
			spring1	0.66	1.85	0.72		
			spring2	6.40	2.40	0.007		
			severity × spring1	0.24	0.32	0.46		
			severity × spring2	-0.79	0.30	0.01		

implications for bird conservation and fire management.

4.1. Heathy-dry forest bird responses to prescribed burn severity

Birds in heathy-dry forests showed little evidence for a prescribed burn severity response. Similar observations have been made elsewhere in south-east Australia. A study of birds after severe wildfire in diverse foothill forests of the Great Dividing Range in south-eastern Victoria found that birds were 77% less numerous in the first winter post-fire, and some had redistributed across the landscape by the first spring (Loyn and McNabb, 2015). In our study, burns were in autumn (March, April) and the bird observations were collected during the first and second spring post-fire. Our results indicated that assemblages had recovered; we observed a rapid recovery by most species, regardless of

the degree of prescribed-burn severity.

The high-severity prescribed burns in this study removed almost 100% of the ground cover and canopy layers (Fig. 2), (Supplementary S2). It might be expected that the significant structural changes would result in a short-term drop in species richness. However, there was no evidence of this occurring. Other studies have found that many species decline in abundance soon after severe bushfires (Loyn, 1997, Loyn and McNabb, 2015). However, in this study, most of the subsequent declines in the occurrence and abundance of individual species were reflected in both the burnt and unburnt sites. The muted response to fire severity by birds in the heathy-dry forest was different to that seen in other forests, for example, the Canadian boreal forests. In these vegetation biomes, fire severity is crucial to bird occurrence. Burnt boreal forests provide habitat for many deadwood-dependent species (Hutto,

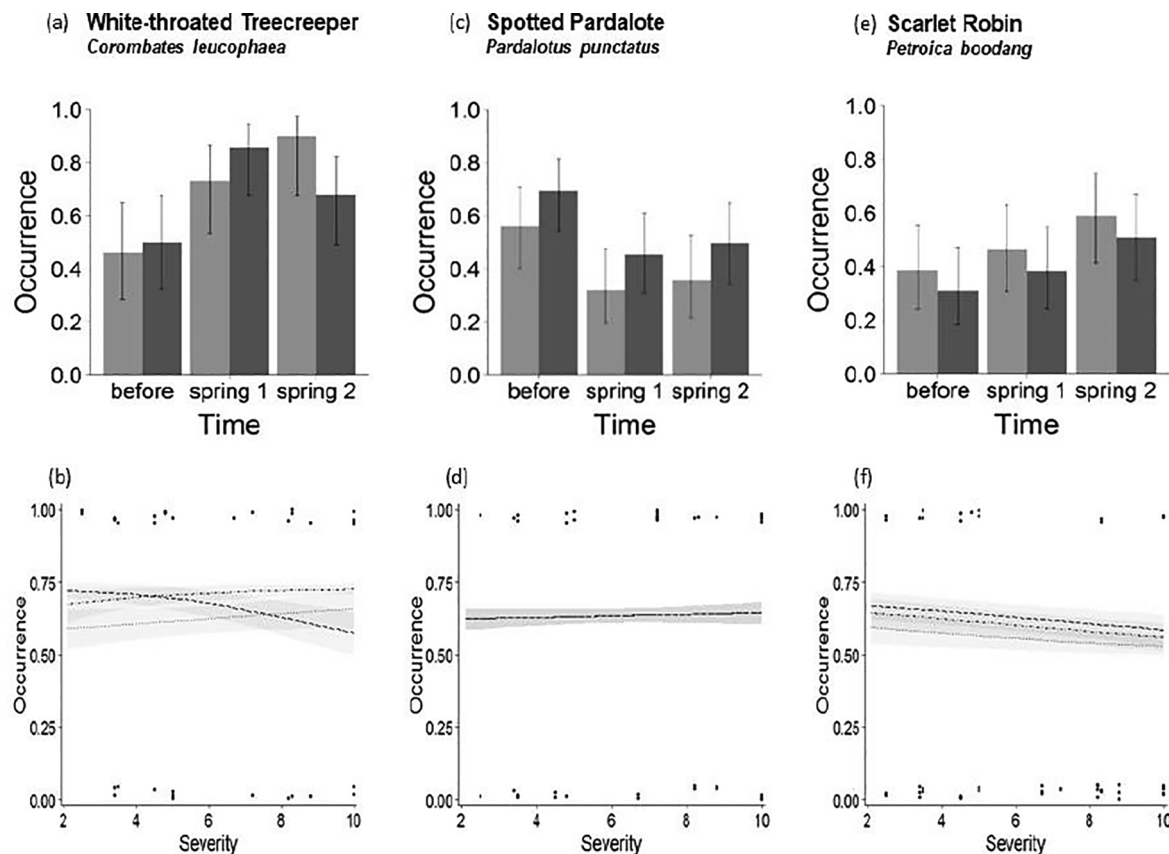


Fig. 4. Individual species' responses to fire and the severity of prescribed burns. Fire graphs (a, c, e) have pairs of control sites (light grey) with burn sites (dark grey) for time periods before the autumn prescribed burns (before) followed by the first observation year (spring1) and the second observation year (spring2). Severity responses are modelled (b, d, f) with continuous data, showing raw data spread across the severity scale. The fixed effect of time is illustrated with a baseline for before (dotted line) on the burnt sites with comparisons for the two years post fire: spring1 (dot and dash) and spring2 (long dash). Graphs show predicted mean effects from generalized linear mixed models and grey shading/errors bars represent 95% confidence limits.

1995, Nappi and Drapeau, 2009), and so the severity of a fire determines the responses of the deadwood-associated insects and birds.

Given the widely-recognised impact of fire on many birds, it was predicted that there would be an increase in species turnover immediately post-fire, with turnover returning to pre-burn levels by the second year. Again, however, there was no evidence, in this study, of this occurring. This result aligns with those from other south-east Australian studies that indicate a scarcity of early-successional stage species in eucalypt forests and woodlands (Loyn, 1980, Lindenmayer et al., 2008, Lindenmayer et al., 2014, Sitters et al., 2014a, Loyn and McNabb, 2015, Sitters et al., 2015). So, there may be a paucity of species across a heathy-dry forest landscape that enter the forest soon after a fire to drive the expected increase in turnover.

The low species turnover in the heathy-dry forest may also be due to rapid vegetation regrowth that occurs in south-east Australian forests (Loyn, 1997, Lindenmayer et al., 2008), which are distinctly different from forest characterised by slower regrowth post-fire, such as occurs in North American conifer forests (Franklin et al., 2002). Our sites developed dense epicormic trunk growth on the severely burnt sites within six months post-fire. It was evident that this regrowth provided shelter, foraging and food resources for birds. Other studies have yielded similar results; where species abundances can be directly related to vegetation density (Monamy and Fox, 2000). In the United States, the conifer-dominated forests have a post-fire landscape with a distinct group of early successional stage bird species (Kotliar et al., 2007, Smucker et al., 2005, Hutto et al., 2015). While distinct post-fire assemblages have been observed in the mulga woodland in the central Australian arid zone (Leavesley et al., 2010), there was no evidence for this occurring in the heathy-dry forest.

Lack of species turnover may have reflected the proximity of survey sites to the edges of burns. A number of the survey sites were within 100 m of the burn edge, which, in many cases, was a narrow, unsealed road. Furthermore, burn areas within the landscape were relatively small. Therefore, birds from monitored sites had little distance to fly to escape the fire and readily returned, to vegetation that had rapidly regrown. Fire analyses however, did not show an influx of birds into control sites post-fire. Edge effects may have also influenced these results. Although edge communities are complex, edge responses can be predictable and consistent (Ries et al., 2004). Abrupt anthropogenic edges, such as those generated by severe prescribed burns, have been shown to result in a decline in the abundance of some birds (Comfort et al., 2016), with bird responses at edges varying relative to vegetation density (McWethy et al., 2009). So, as surveys may have been detecting birds attracted to burn edges, associations with the nature of the burn need to be made with care. This is particularly so given that three of the ten species individually assessed (White-throated Treecreeper, Grey Shrike-thrush and Grey Fantail) have been found to be more abundant in edge sites (Berry, 2001). More generally, fire edges may have important influences on faunal movement, resource selection and biotic interactions (Parkins et al., 2018), but additional research is needed to determine these effects.

In this study, fire size may have influenced bird responses. A comprehensive assessment of pre- and post-fire bird responses to the conflagration of wildfires that burned 450,000 ha across Victoria in 2009, revealed strong evidence of the effects of landscape-level fire severity and fire extent on bird species richness (Lindenmayer et al., 2014). While there are clearly many differences between a whole-of-landscape wildfire and a controlled prescribed burn in terms of fuel loads, and fire

weather, most birds appeared resilient to an increase in the severity of small prescribed burns in heathy-dry forests.

The degree of canopy burn may be a major determinant of bird responses to fire severity (Lindenmayer et al., 2014, Haslem et al., 2016). The primary way fire affects vegetation structure is via attributes such as severity, so low severity forest fires rarely affect canopy strata (Haslem et al., 2016), and most species persist when the overstorey remains intact (Lindenmayer et al., 2014). However, in the heathy-dry forest, both alpha and beta diversity comparisons, between pre- and post-fire bird occurrences were unchanged, regardless of the severity gradient.

For a more comprehensive analysis, it would be necessary to collect more bird data (either spatially or temporally) to allow for a more robust analysis based on abundance models. When species are rare, there is little additional variation in abundance data than in occurrence data alone. For more common species, abundance data contain more information and will, therefore, reveal more about population dynamics than models based on mere presence or absence (Kelly et al., 2017).

4.2. White-throated Treecreeper response to an increase in burn severity

The one species that showed a response to fire and the severity of burns was the White-throated Treecreeper. In the second spring after fire the treecreeper's occurrence declined with increasing burn severity. As the White-throated Treecreeper is a sedentary bird that roosts and nests in tree hollows (Loyn et al., 2007), its requirement for a supply of hollows may not have been met. Hence, its response to fire may be partly a consequence of its needs for roosting and nesting, as well as food. This species has been found to maintain abundance immediately post wildfire in one study in East Gippsland (Loyn, 1997) while decreasing somewhat after the more severe Black Saturday (2009) wildfire in Bunyip State Park (Loyn and McNabb, 2015). While the severity of the most recent fire will strongly influence key resources (Haslem et al., 2016), it is likely that severe fires will have greater effects than moderate fires on both food and sites for roosting and nesting.

The White-throated Treecreeper normally has a preference for feeding from rough-barked trees (Noske, 1985), and rough barked trees retain their rough bark (and presumably much of the bark-dwelling invertebrate fauna) even when outer layers are burned in severe fire. These results accord with another study that showed that the White-throated Treecreeper declined post wildfire, albeit to a much lesser extent than other species (Loyn and McNabb, 2015).

The treecreepers' response may have been exacerbated by drought conditions in the region. Research on ground-dwelling mammals has flagged that abundances may drop in the years following fire, as drought conditions persist (Recher et al., 2009).

4.3. BACI design controls for environmental factors other than fire

Lack of rainfall dries the landscape and the vegetation, increasing dry fuel loads (Bradstock, 2010). In this project, fire and severity analyses flagged that post-fire changes were reflected similarly across burnt and unburnt sites. This aligns with a decrease in rainfall in those years. Even though rainfall in 2010 and 2011 marked the end of the Millennium Drought across regions of Australia, the western and central regions continued drying through 2012 to 2014 (Bureau of Meteorology, 2017). As the monitoring of birds post-fire occurred across this time-period, the drop in numbers across both burnt and unburnt sites may be influenced by drought conditions. Successful bird breeding operates within a range largely driven by two factors - temperature and precipitation (Baker, 1939, James and Shugart, 1974). We did not address questions directly related to changes in rainfall, but it was clear that changes in bird numbers corresponded with substantial decreases in rainfall across the region. This illustrates that, even though the heathy-dry forest birds were resilient to small-scale prescribed burns of any severity, overall abundances may have been in decline. It is understood

that annual fluctuations in rainfall can have dramatic and immediate effects on bird breeding (Gibbs et al., 2011). Longer-term studies, that can more clearly capture short term pulses in breeding, along with the longer-term trends and fluctuations, are needed to partition the effects of fire regimes under a variable and changing climate.

5. Conclusion

More of the landscape across Australia is being burnt to a greater extent (Fairman et al., 2016) and yet, there are insufficient data to determine if current prescribed burn practices have been successful in meeting multiple objectives (Penman et al., 2011). Our findings support the notion that, for small prescribed burns applied in extensive forest patches, bird numbers rapidly rebound to pre-fire levels. However, the study also shows that increasing the severity level of burns may negatively affect some birds, such as hollow-nesters, that have specific requirements that may be reduced or unavailable in a severely-burnt landscape.

CRediT authorship contribution statement

Diana Kuchinke: Conceptualization, Investigation, Methodology, Formal analysis, Writing - original draft, Writing - review & editing. **Julian Stefano:** Methodology, Formal analysis, Writing - review & editing. **Holly Sitters:** Methodology, Formal analysis, Writing - review & editing. **Richard Loyn:** Writing - review & editing. **Peter Gell:** Supervision, Writing - review & editing. **Grant Palmer:** Investigation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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